Low-Noise CMOS LDO Regulator with Bandgap Reference

On-chip ADC supply regulator | GPDK 180 nm | Cadence Virtuoso | 1.8 V → 1.6 V

Overview

This project is a low-noise, low-dropout (LDO) voltage regulator designed in the GPDK 180 nm CMOS process to supply an on-chip ADC. A folded-cascode error amplifier drives a large PMOS pass transistor, and a temperature-stable bandgap reference (BGR) provides the 1 V reference. The regulator generates a 1.6 V output from a 1.8 V supply (200 mV dropout) with high loop gain and a phase margin of about 90° for stable operation.

Key Specifications

Parameter	Value
Technology	GPDK 180 nm CMOS
Supply voltage	1.8 V
Regulated output	1.6 V (design target)
Dropout voltage	200 mV
Reference voltage	1 V (from bandgap reference)
Loop gain (DC)	62.5 dB (ideal 1 V reference); 68.0 dB (with BGR)
Unity-gain frequency	≈ 4.56 MHz
Phase margin	≈ 90° (phase at unity-gain ≈ −92°)
Error amplifier	Folded cascode; 48.1 dB gain, 3.12 MHz UGF, 89.7° phase
Pass device	PMOS, W = 8 mm, L = 360 nm
Load capacitor	100 pF
Load step	10 mA → 50 mA (1 µs edges)
Load-step undershoot / overshoot	≈ 41 mV / ≈ 44 mV
PSRR (VDD)	54.8 dB at low frequency; 32.2 dB at 79 kHz
BGR temperature range	−50 °C to 125 °C
BGR output / drift	≈ 1.079–1.092 V, ≈ 65 ppm/°C (box method)

Architecture

The regulator is a standard PMOS-pass LDO in a negative-feedback loop:

- **Bandgap reference:** generates a temperature-stable reference voltage that sets the output.
- **Folded-cascode error amplifier:** compares the reference (Vp) with the divided output (Vm) and provides high DC gain.
- **PMOS pass transistor:** driven by the error amplifier output; delivers the load current from VIN.

- **Resistive feedback divider:** R0 = 60 kΩ (top) and R2 = 100 kΩ (bottom) set the output voltage.

The output voltage is set by the divider ratio:

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Vreg = Vref × (1 + R0 / R2) = 1 V × (1 + 60k / 100k) = 1.6 V
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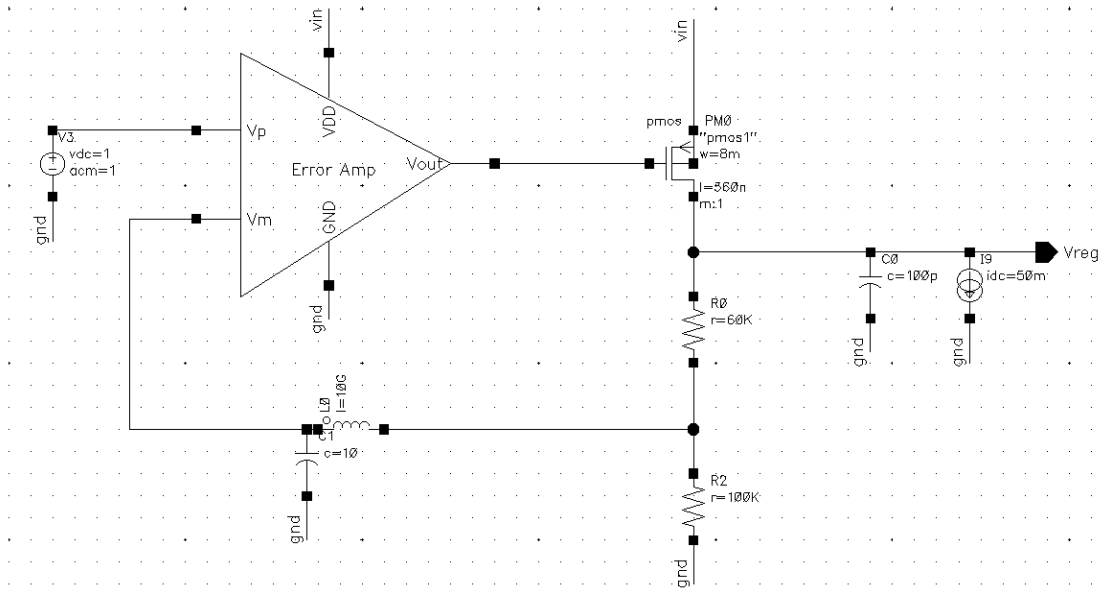


Figure 1. Top-level LDO schematic: error amplifier, PMOS pass device, feedback divider, 100 pF load capacitor and 50 mA load source.

Circuit Design

1. Bandgap Reference

The BGR uses a current-mirror (PMOS/NMOS) structure with bipolar-type diodes in a 1:8 area ratio. The PTAT current generated across R7 (1 kΩ) is summed with the diode voltage across R6 (8.89 kΩ) so that the positive and negative temperature coefficients cancel, giving a first-order temperature-independent output.

- Devices: PM0–PM2 (W = 2 µm, L = 180 nm) and NM0–NM1 (W = 2 µm, L = 180 nm)
- Diodes: D0 and D2 (area 360 f, m = 1), D1 (area 360 f, m = 8)
- Resistors: R7 = 1 kΩ (PTAT), R6 = 8.89 kΩ (output branch)

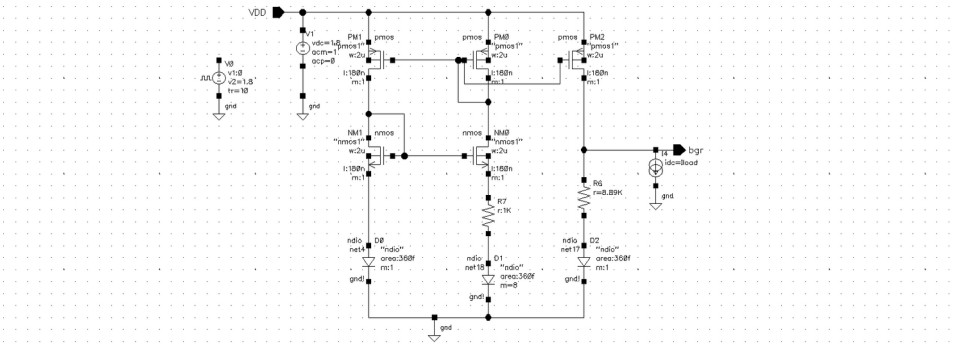


Figure 2. Bandgap reference schematic.

2. Bias Generation

A self-biased beta-multiplier-style core generates the PMOS bias (Vbiasp) at roughly 10 µA per branch. A second bias network derives the cascode bias voltages (Vbias1–Vbias4, Vpcas, Vncas, Vhigh, Vlow) used by the folded-cascode error amplifier.

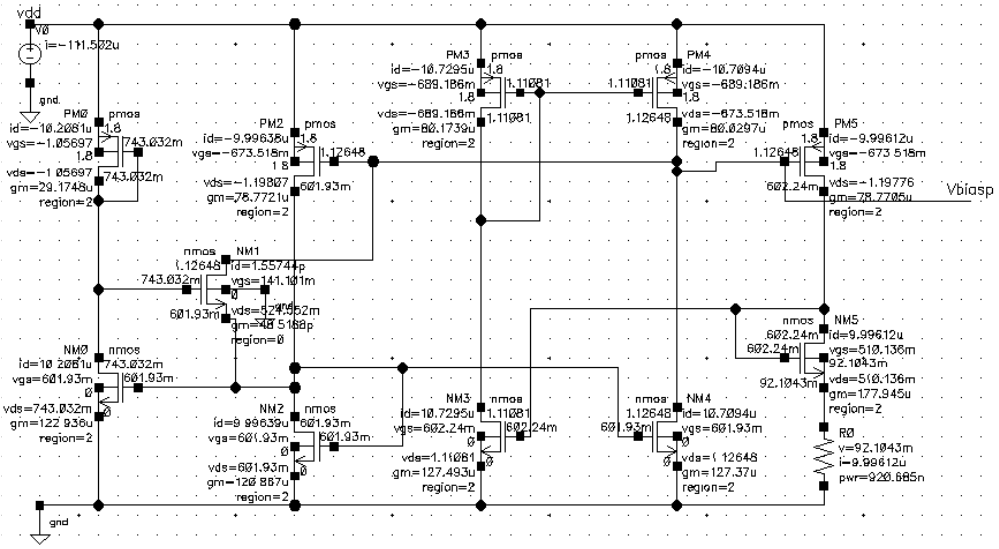


Figure 3. Self-biased current/bias generator (DC operating point annotated).

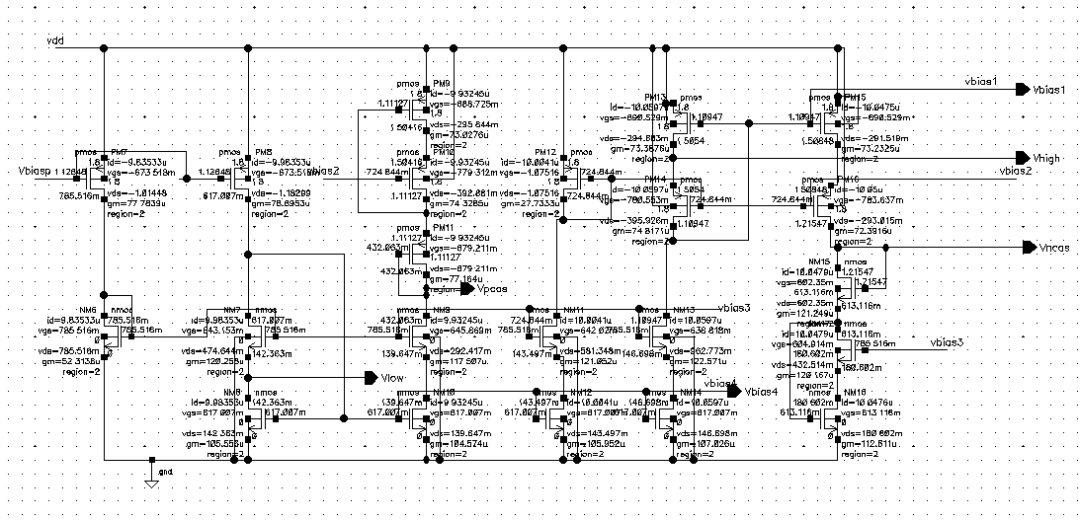


Figure 4. Cascode bias network (DC operating point annotated; all devices in saturation, region = 2).

3. Folded-Cascode Error Amplifier

The error amplifier uses an NMOS differential input pair (W = 1.08 µm, L = 360 nm) with a folded-cascode output stage. This gives high gain in a single stage and good output swing to drive the large gate capacitance of the PMOS pass device. The amplifier is biased from the network above.

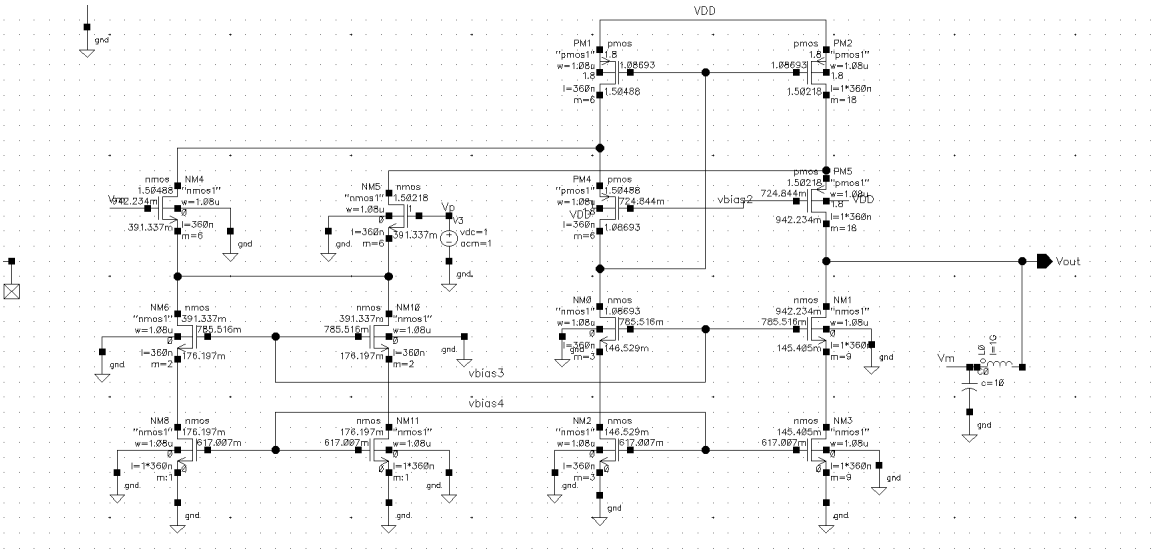


Figure 5. Folded-cascode error amplifier schematic.

4. Pass Transistor and Feedback

The PMOS pass device (W = 8 mm, L = 360 nm) is sized to deliver the 50 mA maximum load current with the 200 mV dropout. A 100 pF output capacitor and the 60 kΩ / 100 kΩ divider complete the regulation loop.

Simulation Results

Bandgap Reference: Temperature Sweep

The BGR output stays within about 12 mV (≈ 1.079 V to 1.092 V) across −50 °C to 125 °C, which corresponds to roughly 65 ppm/°C.

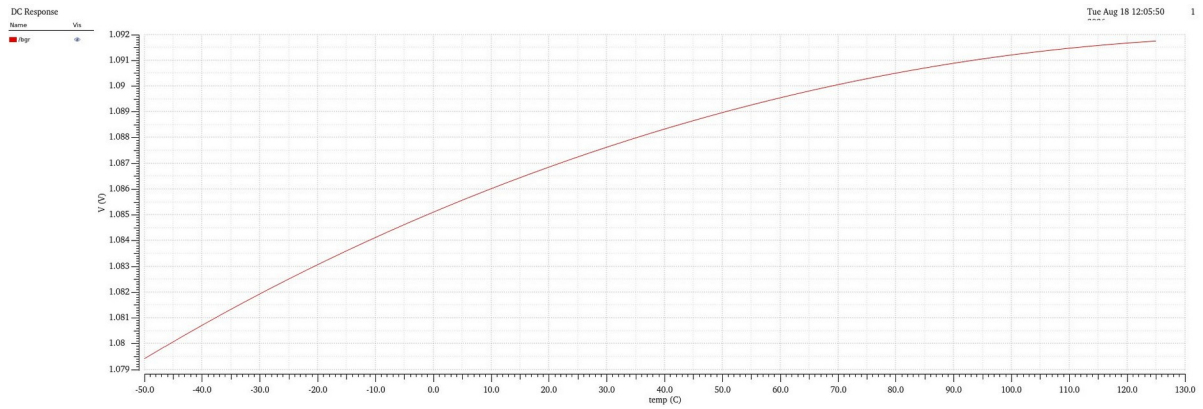


Figure 6. BGR output vs. temperature (−50 °C to 125 °C).

Error Amplifier: Open-Loop Gain and Phase

The folded-cascode amplifier achieves about 48 dB of DC gain with a unity-gain frequency of 3.12 MHz and about 89.7° of phase at crossover.

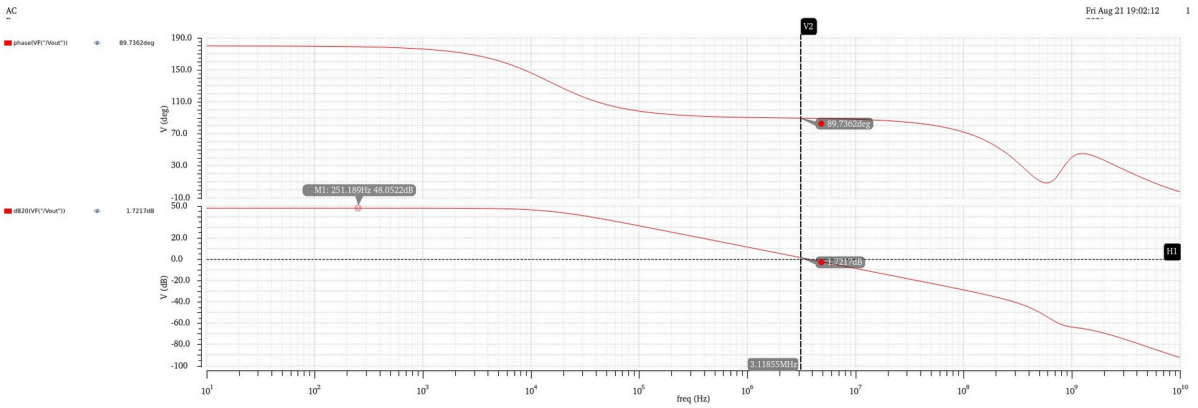


Figure 7. Error amplifier open-loop gain and phase.

LDO Loop Stability

With an ideal 1 V reference the loop gain is 62.5 dB with unity-gain frequency of 4.56 MHz and phase of −92.8° at crossover. With the BGR connected, the DC loop gain is 68.0 dB at the same ≈ 4.56 MHz crossover with −92.1° phase. Both cases leave a phase margin of about 90°.

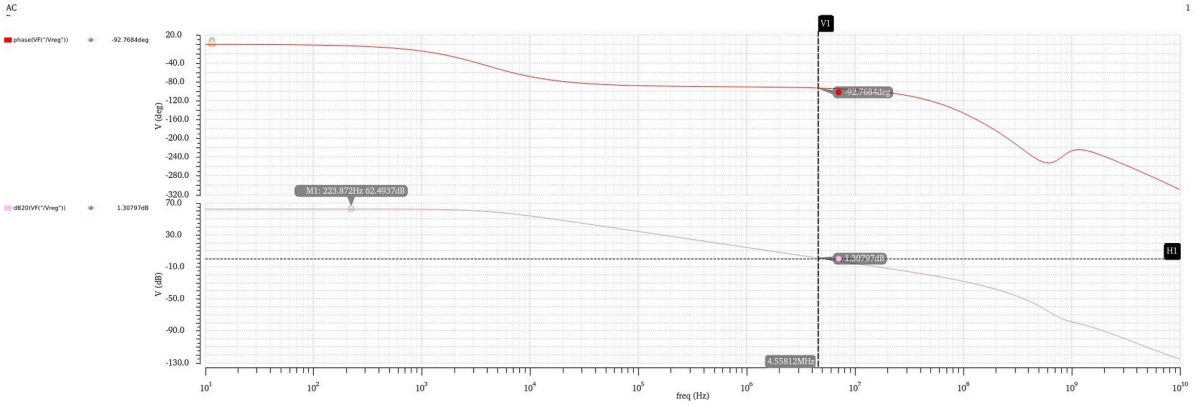


Figure 8. LDO loop gain and phase (ideal 1 V reference).

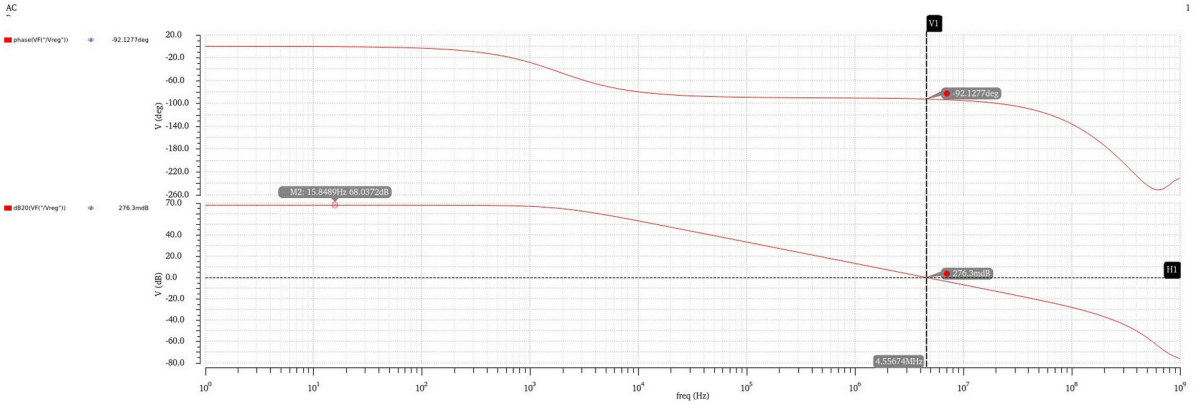


Figure 9. LDO loop gain and phase with the bandgap reference.

Load Transient Response

The load current steps from 10 mA to 50 mA with 1 µs edges (5–6 µs rising, 11–12 µs falling). The output undershoots by about 41 mV and recovers in about 1.1 µs; on the falling edge it overshoots by about 44 mV and settles shortly after.

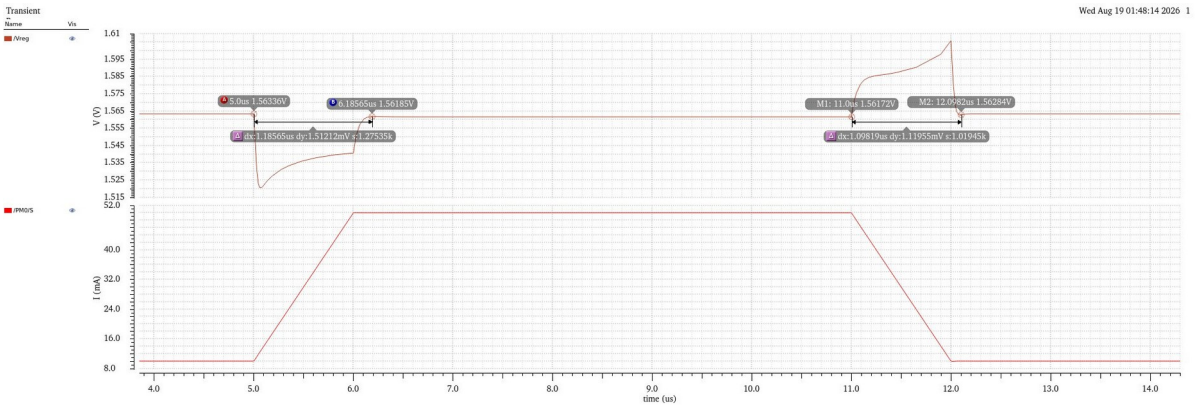


Figure 10. Load transient: Vreg and load current (settling markers).

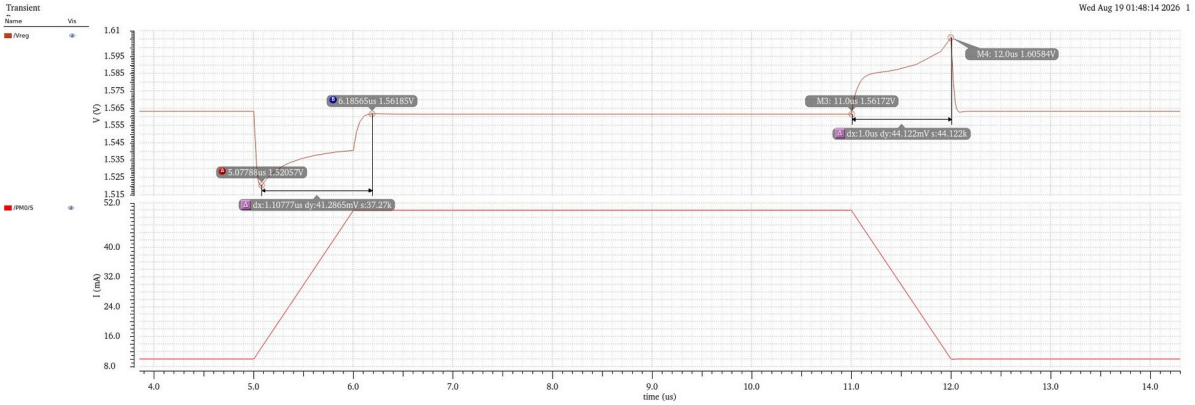


Figure 11. Load transient: peak undershoot (1.5206 V) and overshoot (1.6058 V).

Power-Supply Rejection (PSRR)

Supply (VDD) rejection is 54.8 dB at low frequency and falls to 32.2 dB at 79 kHz, approaching 0 dB near 20 MHz. For ground-side noise, the Vreg response is −2.8 dB at low frequency with a notch near 17.8 MHz (about 22 dB of rejection).

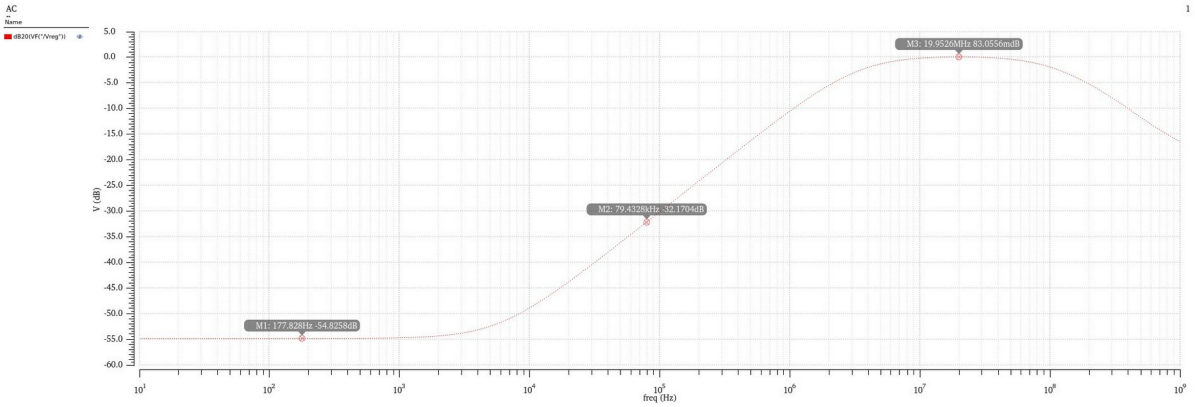


Figure 12. PSRR from VDD to Vreg.

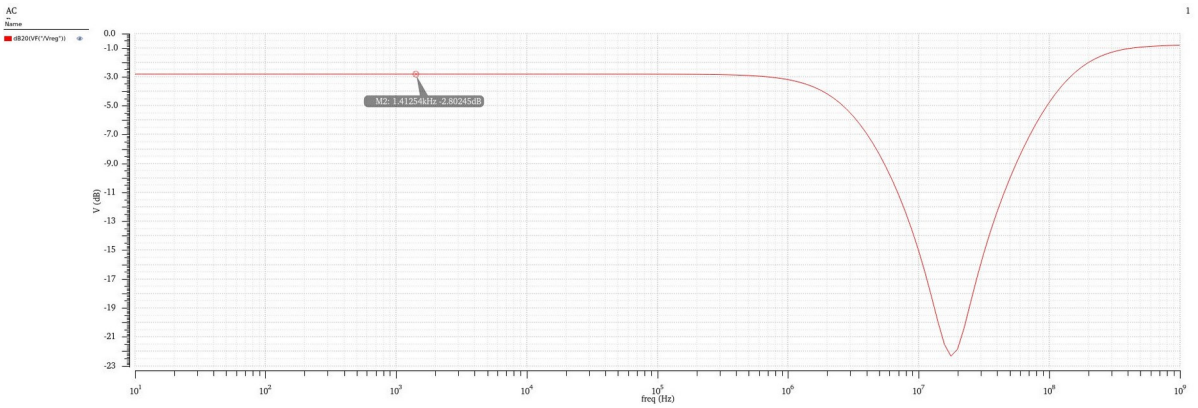


Figure 13. Ground-to-Vreg transfer (dB).

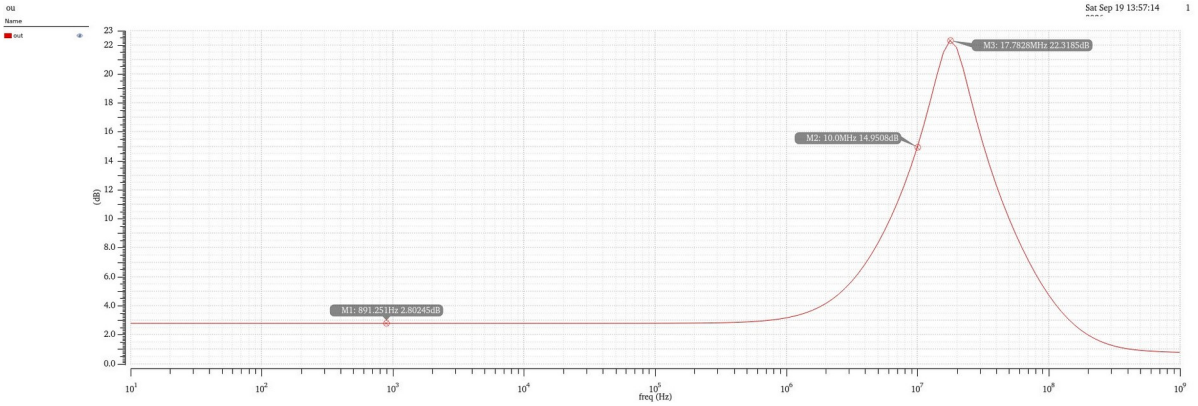


Figure 14. Ground noise rejection (inverted polarity of Figure 13).

Switching-Noise Test

A 1.8 V pulse train (about 20 ns period) is injected and the resulting ripple on Vreg is measured at roughly 69 mV peak-to-peak.

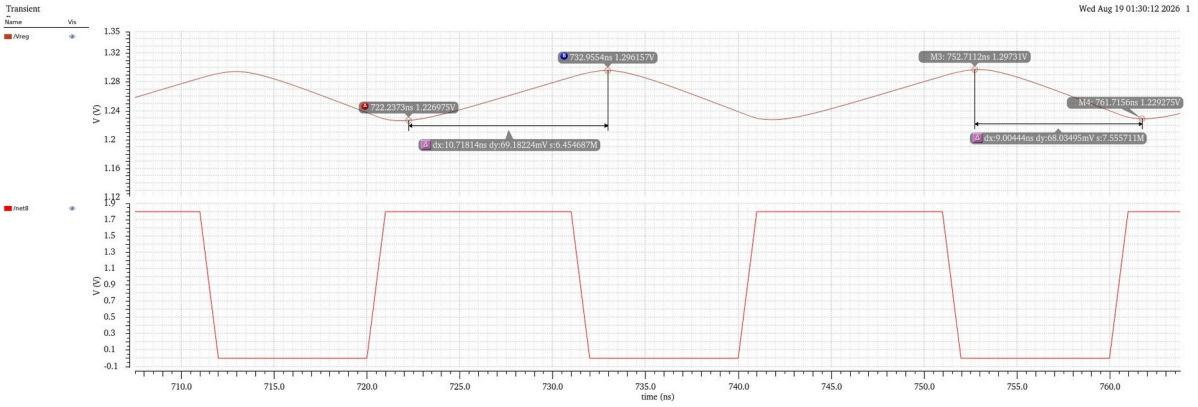


Figure 15. Vreg response to the pulsed disturbance.

Tools and Setup

- Cadence Virtuoso (schematic capture) and Spectre / ADE for simulation
- GPDK 180 nm PDK (1.8 V devices)
- Analyses used: DC temperature sweep, AC (open-loop gain/phase, PSRR), and transient (load step, supply disturbance)